

RITESH KUMAR

riteshraj339955@gmail.com | Greater Noida | [LinkedIn](#) | [GitHub](#)

Computer Science undergraduate with 4+ years of hands-on coding experience and a strong focus on Artificial Intelligence and Machine Learning. Experienced in building end-to-end machine learning solutions, including predictive analytics, recommendation systems, and intelligent web applications using Python, Scikit-Learn, Flask, and related technologies. Eager to apply technical and problem-solving skills to real-world AI-driven challenges.

WORK EXPERIENCE

Generative AI Intern – Google Cloud (Virtual)

Sep 2024 – Oct 2024

- Developed GeminiDecode, a Streamlit-based multilingual document analysis application using Google Gemini APIs for content extraction, translation, and AI-powered insights.
- Integrated Generative AI models with Python applications for automate document understanding.
- Tech Stack:** Python, Streamlit, Google Gemini API, Pillow (PIL), dotenv

AI Intern – Green Tech AI – IBM (AICTE, Shell India & Edunet)

Apr 2025 – May 2025

- Applied AI/ML techniques to sustainability and energy-efficiency use cases using real-world datasets.
- Developed data-driven solution concepts for environmental and green technology challenges.
- Tech Stack:** Python, Pandas, NumPy, Scikit-Learn, Data Analysis, Machine Learning

Cybersecurity Intern – IBM (AICTE & Edunet – SkillsBuild)

Jan 2025 – Feb 2025

- Built an image steganography application using LSB-based data hiding techniques.
- Explored cybersecurity concepts including information security, threat detection, secure communication, and system protection principles.
- Tech Stack:** Python, Image Processing, Steganography, LSB Encoding

PROJECTS

Plant Disease Detection System

- Developed a Computer Vision-based plant disease detection system using a Scikit-Learn classification pipeline (StandardScaler + SGDClassifier) trained on handcrafted image features (color histograms, texture, and edge statistics).
- Trained and validated the model on 87,800+ leaf images (70,295 training / 17,572 validation) across 38 disease and healthy plant classes, achieving 84.77% validation accuracy.
- Applied image preprocessing and augmentation techniques to improve model performance and generalization.
- Deployed the model behind a Flask REST API with a React and Tailwind CSS frontend, delivering real-time disease prediction with confidence scores and treatment guidance from uploaded leaf images.
- Tech Stack: Python, Flask, Scikit-Learn, NumPy, Pillow, React, Tailwind CSS
- GitHub: <https://github.com/RIT8603/Plant-Disease-Detection-System>
- Live Demo: <https://plant-disease-frontend-9zza.onrender.com/>

Book Recommendation System

- Developed a personalized book recommendation system using Machine Learning techniques and user preference analysis.
- Processed and analyzed book ratings and user interaction data to generate relevant recommendations.
- Implemented popularity-based and item-similarity collaborative filtering on the Book-Crossing dataset, filtering to books with 250+ ratings and users with 200+ ratings to ensure reliable recommendations.
- Built a Flask web application with a Supabase-backed admin dashboard for managing book data and retraining the recommendation model on demand.
- GitHub: <https://github.com/RIT8603/Book-Recommend-System>
- Live Demo: <https://book-recommend-system-blsz.onrender.com/recommend>

Customer Churn Prediction

- Developed a Machine Learning model to predict customer churn and identify customers at risk of leaving a service.
- Performed data preprocessing and EDA, and engineered features including tenure grouping and average spend per tenure period to strengthen model signal on customer behavioral data.
- Trained and compared five classification algorithms (Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost) on the Telco Customer Churn dataset (7,043 customers); selected Logistic Regression as the best performer with 80.48% accuracy, 0.846 ROC-AUC, and 0.595 F1-score.
- Built an interactive web application for real-time churn prediction and business insights.
- GitHub: https://github.com/RIT8603/churn_prediction
- Live Demo: <https://churn-prediction-3ro1.onrender.com/>

EDUCATION

B.Tech CSE (Lateral Entry), MIET Noida | 2025–2028

Diploma CSE, Government Polytechnic Barauni, Begusarai | 2022–2025

Programming

- Python, C++, SQL

Data Analysis

- Pandas, NumPy, Matplotlib

Machine Learning

- Scikit-Learn, TensorFlow, OpenCV

Web & Deployment

- Flask, Streamlit, React, Tailwind CSS,
- AI-Assisted Web Development

Tools

- Git, GitHub, Jupyter Notebook, Vs Code
- Colab, AWS, Render, Vercel, Kaggle, Codex, GitHub Copilot

AI & Generative AI

- Prompt Engineering
- Google Gemini API

LANGUAGES

- English
- Hindi
- Bhojpuri
- Tamil

CERTIFICATIONS

- Mastering Machine Learning:** From Beginner to Expert – GeeksforGeeks (2026)
- Introduction to Firewall** – Great Learning Academy (2023)
- Blockchain Basics** – Great Learning Academy (2023)

ACHIEVEMENTS

- Participated in 5+ hackathons, including IIT Patna Hackathon and Smart India Hackathon (SIH), solving real-world challenges under strict deadlines.
- Served as a College Campus Mentor; organized multiple technical events for the student community.
- Member, Google Developer Group (GDG) Patna.
- Admin of 5+ tech-focused community groups, collectively managing 1,000+ students.

HOBBIES

- Playing chess and carrom
- Reading books (technology and personal growth)
- Cycling
- Listening to music